

FiNER-ORD: Financial Named Entity Recognition Open Research Dataset

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Abstract

Over the last two decades, the development of the CoNLL-2003 named entity recognition (NER) dataset has helped enhance the capabilities of deep learning and natural language processing (NLP). The finance domain, characterized by its unique semantic and lexical variations for the same entities, presents specific challenges to the NER task; thus, a domain-specific customized dataset is crucial for advancing research in this field. In our work, we develop the first high-quality English Financial NER Open Research Dataset (FiNER-ORD). We benchmark multiple pre-trained language models (PLMs) and large-language models (LLMs) on FiNER-ORD. We believe our proposed FiNER-ORD dataset will open future opportunities to use FiNER-ORD as a benchmark for financial domain-specific NER and NLP tasks. Our dataset, models, and code are publicly available on GitHub and Hugging Face under CC BY-NC 4.0 license.

1 Introduction

The growth of technology and the web over the last several decades has led to a rapid increase in the generation of text data, especially in the financial domain. The abundance of financial texts, primarily in the form of news and regulatory filings, presents a valuable resource for analysts, researchers, and even individual investors to extract relevant information. With the given abundance of text data, manually extracting relevant information is impossible and unsustainable for large-scale text datasets used for rapid downstream tasks. However, NLP techniques can help with automating the information retrieval process. Named entity recognition (NER) is one such NLP task that serves as an important first step to identify named entities, such as persons, organizations, and locations, and efficiently use available text data to generate a knowledge graph.

While numerous studies have constructed annotated datasets (Sang and De Meulder, 2003; Derczynski et al., 2017a; Weischedel et al., 2013) and developed NER models (Li et al., 2019; Yamada et al., 2020; Wang et al., 2021a,b) for generic texts, the financial domain presents unique challenges that necessitate the use of domain-specific text. The current annotated dataset based on Credit Risk Agreements (Alvarado et al., 2015) for financial NER has significant shortcomings, limiting the use of deep learning models for financial NER. We elaborate on these shortcomings in Section 3.2.

In addressing these challenges within the scope of data customization and ethics-conscious evaluation, our work contributes significantly to the field by establishing the largest Financial Named Entity Recognition Open Research Dataset (FiNER-ORD). Our objective in this context is to create an evaluation dataset and evaluate existing generalist, non-customized models. We evaluate pre-trained language models (PLMs) and zero-shot large-language models (LLMs) on the FiNER-ORD dataset.

Jimmy Cao **PER**, a Beijing **LOC**-based **BMW** **ORG** spokesman, said he was not able to provide an immediate comment.

The average “maker suggested retail price” (MSRP) for all passenger cars remains relatively high in China **LOC**, at around 280,000 yuan (\$45,000), according to research firm JATO Dynamics **ORG**.

Figure 1: Representative example of annotation in FiNER-ORD.

2 Related Work

Named Entity Recognition Datasets Commonly used general NER datasets include the CoNLL-2003 (Sang and De Meulder, 2003),

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OntoNotes5.0 (Weischedel et al., 2013), WNUT 2017 (Derczynski et al., 2017b), and FewNERD (Ding et al., 2021). FiNER-139 from Loukas et al. (2022) is an entity recognition dataset for numerical financial data. Like our work, they also don’t use numerical entities because they can be easily extracted using regular expressions and are the easiest entity types to recognize. There has also been work in recognizing entities in invoices, business forms, and emails (Francis et al., 2019). CRA NER dataset from Alvarado et al. (2015) was the first attempt to create a financial NER dataset similar to FiNER-ORD, but in our assessment presented in Section 3.2, it is skewed and thus limited in usefulness.

Applications for Information Retrieval in Finance Sentiment analysis of news aids event ranking based on market factors, aiding price prediction (Feng et al., 2021). Analyzing tweets and headlines automates trading, price movement, and risk forecasts (Sawhney et al., 2021). Extracting claims from analysts’ reports improves volatility forecasting on release and earning dates (Shah et al., 2024). Real-time web tools extract operating segments, aiding performance analysis (Ma et al., 2020). Transformer models fine-tuned for price-change data extraction measure inflation exposure (Chava et al., 2022). Financial knowledge graphs from news data enhance algorithmic trading (Cheng et al., 2020).

3 Dataset

3.1 FiNER-ORD

FiNER-ORD consists of a manually annotated dataset of financial news articles (in English) collected from webz.io¹. In total, there are 47,851 news articles available for download in this dataset at the time of writing this paper.

Sampling and Manual Annotation For the manual annotation of named entities in financial news, we randomly sampled 220 documents from the entire set of news articles. We observed that some articles were empty in our sample, so after filtering the empty documents, we were left with a total of 201 articles. We use the open-source Doccano² annotation tool (see Appendix G for more details) to ingest the raw dataset and manually label person

FiNER-ORD	Train	Validation	Test
Articles	135	24	42
Sentences	3,262	402	1,075
Tokens	80,531	10,233	25,957
PER	821	138	284
LOC	966	193	300
ORG	2,026	274	544

Table 1: Number of articles, sentences, tokens, and entities (person, location, organization) in the train, validation, and test splits of FiNER-ORD.

(PER), location (LOC), and organization (ORG) entities. Figure 1 shows an example of manually annotated named entities in FiNER-ORD. Labeling was performed independently by two different annotators, following the predefined annotation guide presented in Appendix E, to reduce potential labeling bias. Annotator labeling was compared and validated for consistency. We provide details about annotator background in Appendix B, annotator agreement in Appendix C, and post-processing in Appendix D.

Dataset Statistics Following the manual annotation and post-processing procedures, each token is labeled as one of 4 broad entity types: PER, LOC, ORG, and O. As discussed in Appendix D each of the PER, LOC, and ORG classes are further segmented with the suffixes `_B` (denoting beginning token of a span) and `_I` (denoting intermediate token of a multi-token span), respectively. As discussed earlier, we manually annotate the train, validation, and test splits of the 201 articles in FiNER-ORD. The descriptive statistics on the resulting FiNER-ORD are available in Table 1.

3.2 Comparison of Datasets

Credit Risk Agreements (CRA) Table 2 compares our proposed dataset (FiNER-ORD) with the CRA NER dataset (Alvarado et al., 2015). The CRA dataset attempts to provide a domain-specific dataset in CoNLL format using manual annotation on 8 English documents from the U.S. Security and Exchange Commission (SEC) filings (Bird et al., 2009). However, the annotation methodology for the CRA NER dataset "automatically tagged all instances of the tokens *lender* and *borrower* as being of entity type PER" (Alvarado et al., 2015). This approach is problematic because of the resulting skewed distribution of entity types in the dataset, leading to confounded results. Our analysis of the CRA NER dataset showed that in FIN3 (CRA test data split), instances of the tokens *lender* and *bor-*

¹<https://webz.io/free-datasets/financial-news-articles/>

²<https://github.com/doccano/doccano>

rower represented 83.05% of all PER tokens and 44.95% of all tokens labeled as PER, ORG, MISC, or LOC. Similarly, in FIN5 (CRA train data split), instances of the tokens *lender* and *borrower* represented 90.04% of all PER tokens and 46.08% of all tokens labeled as PER, ORG, MISC, or LOC.

Thus, we believe the CRA dataset is not a high-quality benchmark for specialized NLP tasks in the financial domain, motivating us to create a new high-quality financial domain-specific NER dataset. The dataset comparison statistics are available in Table 2. We also perform a transfer learning study presented in Section 5.1 in order to further emphasize the importance of FiNER-ORD in comparison to the CRA NER dataset.

CoNLL-2003 The CoNLL NER dataset (Sang and De Meulder, 2003) was created with manual annotation on Reuters generic news stories. In general, financial texts differ from general texts and contain a higher ratio of organization tokens and entities compared to person and location tokens and entities. This can be important for financial news network applications that need to capture the interdependence of companies and the complexity of modern financial markets. In the FiNER-ORD dataset, the ratio of organizations (ORG) to location (LOC) to person (PER) entities is 2.29:1.17:1. This distribution contrasts with the CRA and CoNLL datasets, which exhibit ORG:LOC:PER ratios of 0.31:0.22:1 and 0.93:1.06:1, respectively.

We also analyze the average length of each entity-type in terms of the number of tokens per entity. In the order of FiNER-ORD, CRA, and CoNLL, we see the average number of tokens for a PER entity to be approximately 1.66, 1.06, and 1.69, respectively. Similarly, for LOC entities we find an average length of approximately 1.34, 2.09, and 1.50, respectively. For ORG entities, we find an average length of 1.72, 1.69, 1.18, respectively. These findings show that for PER entities, the average length is similar in FiNER-ORD and CoNLL. However, the skewed distribution of PER entities with the problematic labeling of *lender* and *borrower* as PER in CRA leads to an average length value close to 1 for PER entities in CRA, underscoring the finance-specific utility of FiNER-ORD over CRA.

Similarly, for LOC entities, the average length values are similar in FiNER-ORD and CoNLL. However, the finance-specific usefulness of FiNER-

Dataset	FiNER-ORD	CRA	CoNLL
Articles	201	8	1,393
Sentences	4,739	1,473	22,137
Tokens	116,721	54,262	301,418
PER	1,243	962	10,059
LOC	1,459	208	10,645
ORG	2,844	295	9,323

Table 2: Comparison of our FiNER-ORD with Credit Risk Agreements and CoNLL-2003 English NER datasets in terms of the number of articles, sentences, tokens, and entities (person, location, organization).

ORD over CoNLL is highlighted by the average length values for ORG entities. We see similar average lengths for ORG entities in FiNER-ORD and CRA, both of which are financial NER datasets, but the average length of ORG entities in CoNLL is much smaller. Additionally, the comparison between the ratios of ORG_B and ORG_I tokens for FiNER-ORD (1.4:1) and CoNLL (5.6:1) suggests that in financial texts, organization entities are more likely to span over multiple tokens. Having a higher percentage of useful ORG entities and tokens can also enhance applications like measuring correlations in financial news networks and market movements (Wan et al., 2021). This highlights the importance of creating high-quality datasets specifically for the financial domain.

4 Models

4.1 PLMs

We benchmark FiNER-ORD with several base and large transformer-based models. For the base model category, we use BERT (Devlin et al., 2018), FinBERT (Yang et al., 2020), and RoBERTa (Liu et al., 2019). For the large model category, we use BERT-large (Devlin et al., 2018) and RoBERTa-large (Liu et al., 2019). We do not pre-train these models before fine-tuning them. Further details about fine-tuning is provided in Appendix H.

4.2 LLMs

To benchmark the performance of current SOTA generative LLMs, we measure the zero-shot performance on three train test splits for the "gpt-4o-2024-08-06"³, "Meta-Llama-3.1-405B-Instruct-Turbo", and "Meta-Llama-3.1-70B-Instruct-Turbo" models with 0.0 temperature and 1000 max tokens for the output. All API calls were made on August 8, 2024 or August 9, 2024. The zero-shot prompt can be found in Appendix I.

³<https://platform.openai.com/docs/guides/gpt>

notice how all of the pretrained LMs that were fine tuned to the NER task are BERT models

PLM (pre trained Language model)

Model	Train Split	PER	LOC	ORG	Weighted Average
Panel A: FineTuning with PLMs					
BERT-base-cased	FiNER-ORD	0.8811 (0.0192)	0.6820 (0.0239)	0.6013 (0.0490)	0.6931 (0.0327)
FinBERT-base-cased	FiNER-ORD	0.7456 (0.0254)	0.6836 (0.0238)	0.6002 (0.0288)	0.6589 (0.0231)
RoBERTa-base	FiNER-ORD	0.9050 (0.0076)	0.7154 (0.0608)	0.6304 (0.0878)	0.7220 (0.0585)
BERT-large-cased	FiNER-ORD	0.8954 (0.0090)	0.7289 (0.0467)	0.6272 (0.0145)	0.7216 (0.0039)
RoBERTa-large	FiNER-ORD	0.9263 (0.0025)	0.7717 (0.0152)	0.6769 (0.0130)	0.7648 (0.0057)
Panel B: Zero-Shot with Generative LLMs					
Llama-3.1-70B-Turbo	Zero-Shot	0.6706 (0.0033)	0.5919 (0.0027)	0.4981 (0.0038)	0.5661 (0.0004)
Llama-3.1-405B-Turbo	Zero-Shot	0.7413 (0.0020)	0.6587 (0.0017)	0.5751 (0.0024)	0.6389 (0.0016)
GPT-4o	Zero-Shot	0.8004 (0.0065)	0.6651 (0.0073)	0.6036 (0.0111)	0.6692 (0.0057)
Panel C: Transfer Learning Ablation					
RoBERTa-large	CRA	0.5918 (0.0868)	0.0145 (0.0020)	0.0411 (0.0174)	0.1730 (0.0262)
RoBERTa-large	CoNLL	0.8707 (0.0171)	0.7640 (0.0278)	0.5668 (0.0678)	0.6954 (0.0278)

Table 3: Performance comparison of various models tested on the FiNER-ORD test split. Panel A contains results for various models fine-tuned on the FiNER-ORD train sample. Panel B contains results for zero-shot Generative LLMs. Panel C contains results for transfer learning experiments where the RoBERTa-large model is fine-tuned on the Credit Risk Agreements (CRA) NER dataset and CoNLL-2003 dataset. All values are weighted F1 scores. An average of 3 seeds was used for all models. The standard deviation of the F1 scores is reported in parentheses.

5 Results and Analysis

In this section, we evaluate and benchmark different NLP models on the NER task with FiNER-ORD. For all models, we consistently use the same train, validation, and test splits in FiNER-ORD at the sentence-level.

For each PLM, we use three different seeds for the three runs of each model. We run each LLM three times with the same settings discussed in Section 4.2. We report average weighted F1 scores for the best hyper-parameter configuration of each model in Table 3.

Although zero-shot GPT-4o outperforms zero-shot the Llama models, fine-tuned PLMs outperform both zero-shot LLMs across all entity label categories. This finding aligns with the survey by Pikuliak (2023), which finds that zero-shot ChatGPT fails to outperform fine-tuned models on more than 77% of NLP tasks. RoBERTa-base achieves best weighted F1 score overall and for all entity label categories when compared to all models. A future ablation study of these language models would provide key insights such as why RoBERTa-base, a model more generally trained with the masked language modeling (MLM) objective, outperforms (when considering our financial NER evaluation) FinBERT-base, a model trained on financial sentiment classification tasks and widely used for various financial domain-specific NLP tasks. Applying FiNER-ORD as a financial-domain NER benchmark dataset for such studies can further improve customized financial domain-specific language models.

5.1 Transfer Learning Ablation

This study evaluates the transfer learning capabilities of the best-performing model, RoBERTa-large, originally trained and tested on the FiNER-ORD dataset. Specifically, we investigate its performance when fine-tuned on the CRA NER dataset and CoNLL dataset, followed by testing on the test split of the FiNER-ORD dataset. The outcomes of these transfer learning experiments are detailed in Panel C of Table 3. We also evaluate an existing model trained on the CONNL 2003 dataset and test it on FiNER-ORD in Appendix J.2. These results underscore the significance of the FiNER-ORD dataset in enhancing model performance relative to the existing CRA and CoNLL NER datasets.

6 Conclusion

We present the first high-quality, manually annotated financial NER dataset, FiNER-ORD, which was generated from open-source financial news articles. We demonstrate the importance of our FiNER-ORD dataset compared to the existing CoNLL-2003 and financial CRA NER datasets. To evaluate the proposed dataset, we benchmark various configurations of PLMs and LLMs on FiNER-ORD. The performance analysis shows that RoBERTa-base outperforms all tested models (including FinBERT-base) overall, and GPT-4o, Llama-3.1-405B, and Llama-3.1-70B underperform the tested PLMs over all entity label categories. Furthermore, our proposed financial domain-specific dataset and performance analysis together present opportunities to further explore improvements for customized financial domain-specific language models.

Ethics Statement

All language models used, under their respective license categories, are publicly available for our experimental purposes. With regards to ethical considerations for the environmental impact of our experiments, we only fine-tune the benchmarked PLMs because pre-training PLMs are known to have a large carbon footprint. We acknowledge that our dataset is constructed using English news articles and thus biased in terms of language representation and inclusion. The news articles do not contain offensive or discriminatory content. The dataset license and copyright exceptions are stated in Appendix A.

Limitations

We acknowledge a geographic bias in the dataset as FiNER-ORD only includes English financial news articles. In the future, other forms and languages of financial texts could be manually annotated to expand FiNER-ORD. Additionally, more label classes such as ‘product’, ‘miscellaneous’, and forms of relationships could be annotated for future downstream applications such as knowledge graph creation. We do not present a new SOTA model architecture for financial NER and we do not include or discuss finance domain-specific LLMs like BloombergGPT (Wu et al., 2023) in our work as we have no way of accessing it. Our primary focus of this work is to present the first high-quality financial NER open research dataset and benchmark our FiNER-ORD with multiple PLMs and LLMs to evaluate the performance of these models on the finance domain-specific NER task.

Form 10-K Filings When considering data sources for financial NER, we found that SEC Form 10-K filings, filed annually by public companies in the United States, have certain limitations. The 10-K provides details about a company’s business, risks, and operating and financial results for each fiscal year. For the NER task, however, we observe there is a noticeable bias towards organization entity tokens associated with the company filing the 10-K. As a result, there is a reduced diversity of named entities, particularly concerning other organizations. Furthermore, the average length of 10-K filings has been growing significantly every year with at least 60,000 words between 2022 and 2023, a significant increase from around 30,000 words in

2000 and approximately 42,000 words in 2013⁴. In contrast, financial news articles are shorter and can be used to better capture the interdependence of companies and the complexity of modern financial markets. Given this, we decided that constructing a high-quality, manually annotated dataset would be more feasible and information-diverse with the open-source news articles we annotate in FiNER-ORD as opposed to 10-K filings.

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⁴<https://www.wsj.com/articles/the-109-894-word-annual-report-1433203762>

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A Copyright Exceptions

The dataset will be made publicly available on Hugging Face under the non commercial CC BY-NC 4.0 license. The individual articles comprising the dataset can be considered exempt from copyright for non-commercial research. Exceptions to copyright in the United Kingdom are regularly updated at gov.uk⁵. The EU Directive on Copyright in the Digital Single Market⁶ provides exceptions for reproductions made by research organizations.

B Annotator Background

Labeling was performed independently by two different annotators. Annotator 1 is a Doctoral Researcher from India working with NLP and Computational Finance. Annotator 2 is a Masters student from the United States pursuing a Computer Science degree with Machine Learning specialization. Both annotators were male researchers and read English financial news articles daily from several outlets and sources. Additionally, both annotators are authors, so the annotators were not hired or paid for annotations.

⁵<https://www.gov.uk/guidance/exceptions-to-copyright>

⁶<https://eur-lex.europa.eu/eli/dir/2019/790/oj>

C Annotator Agreement

Both annotators had a manual annotation agreement of approximately 96.85% of the 5546 entities across the PER, LOC, ORG entity classes and train, validation, test splits of FiNER-ORD. The annotation guide was referenced to resolve disagreements between the manual labeling of both annotators. The annotation guide was developed iteratively during the annotation process. Online resources were consulted if the annotation guide did not address a specific disagreement, and the annotation guide was updated accordingly afterwards.

D Post-Processing

To correct potential errors in manual annotations, we run a custom post-processing script that performs the following four tasks: (1) remove trailing spaces from annotated entities, (2) extend token-level borders to non-space characters to change an erroneous span to the correct span, (3) clean entity suffixes with techniques such as removing an apostrophe followed by the letter s ('s) from entity suffixes, (4) tokenize text with Stanza (Qi et al., 2020) and add positional information for labeled entities by splitting multi-token spans into separate tokens, assigning _B as the label suffix for the first separated token in the multi-token span, and assigning _I as the label suffix for the remaining separated tokens in the multi-token span. We note that after the post-processing script is run, all tokens which are not annotated with one of PER_B, PER_I, LOC_B, LOC_I, ORG_B, ORG_I are assigned the label O, denoting "other" type of token not belonging to the person, location, organization classes.

E Annotation Guide

The manual annotation process to create FiNER-ORD consisted of ingesting the financial news articles in Doccano. Each news article is available in the form of a JSON document with various metadata information including the source of the article, publication date, author of the article, and the title of the article. Entities of the type *person* (PER), *organization* (ORG), and *location* (LOC) were identified according to the rules described below. Some well-known names for these entities were obvious while others were confirmed by researching the names to identify the correct entity type.

All Entities	PER	LOC	ORG
China-155	Obama-26	China-154	GM-52
U.S.-69	Abbott-19	U.S.-69	Reuters-47
GM-52	Clinton-18	Greece-49	Facebook-35
Greece-49	Bush-14	US-46	Nikkei-34
Reuters-47	Turnbull-14	UK-37	Fed-32
US-46	Varoufakis-11	Australia-32	Apple-31
UK-37	Jaitley-9	London-28	Ford-24
Facebook-35	Shorten-9	Japan-24	Stratasys-23
Nikkei-34	McPhail-9	New York-23	LSE-21
Australia-32	Tony Abbott-8	Europe-22	House-20
Fed-32	Malcolm Turnbull-8	Sydney-21	UK Markets-19
Apple-31	Andrew-8	Taiwan-19	Unilever-19
London-28	Bishop-8	Bahrain-18	Motley Fool-18
Obama-26	John - Erik Koslosky-8	Kenya-18	SEC-18
Japan-24	Glatt-7	India-17	FT-18

Table 4: Most common entities in FiNER-ORD.

E.1 Person Entities

PER entities were identified by their first name and/or last name. In the examples below, bold spans represent a single person entity. In the case where a person was identified by their first and last name, the entire name was labeled as PER and the post-processing script tagged the first name as PER_B and the last name as PER_I. Words like *President*, *Ms*, and *CEO* were not labeled as part of the PER entity but help indicate a PER entity. In a context indicating possession with 's, the name until the 's was tagged.

- President **Obama**
- CEO **Phyllis Wakiaga**
- Ms **Wakiaga**
- **Bill Clinton's**

E.2 Location Entities

LOC entities primarily consisted of names of continents, countries, states, cities, and addresses. In the examples below, bolded spans represent spans comprising LOC entities. Commas in addresses are not included tagged LOC entities. In such cases for tagging addresses, each complete span delimited by a comma was tagged as a LOC entity. In a context indicating possession with 's, the name until the 's was tagged. In the case where a location was identified by multiple tokens delimited by a space, the entire name was labeled as LOC and the post-processing script tagged the first name as LOC_B and the last name as LOC_I. Words such as "Kenyan" were treated as adjectives and thus not labeled as a LOC entity. When discussing a law-maker's political and location affiliation, examples

such as *R-Texas* denoting "Republican from Texas" are encountered, in which only the location name such as *Texas* is tagged.

- **Asia**
- **US**
- **India**
- **United States**
- **Beijing**
- **New York**
- **Redwood City, California**
- **Kenya's**
- **Mombasa Road**
- **R-Texas**

E.3 Organization Entities

ORG entities consist of examples such as company names, news agencies, government entities, and abbreviations such as stock exchange names and company stock tickers. Punctuation marks such as hyphens are included when tagging an ORG entity. As designated, *.com* is included in the identified company's name. In a context indicating possession with 's, the name until the 's was tagged.

- **Wal-Mart**
- **China Resources SZITIC Trust Co Ltd**
- **The Wall Street Journal**
- **Atlanta Federal Reserve**

- **Morgan Stanley’s**
- **Delta Air Lines**
- **DAL**
- **NYSE**
- **Amazon.com**

E.4 Annotation Edge Cases

There were a few edge cases in the annotation process. For example, labeling entities when location (LOC) is part of the organization (ORG) entity is a common problem in finance. The phrase "Google India" has "India" which is a location, but it is labeled as an organization in our framework. This is because our process does not permit overlapping entity labels. Another such example is "New York Stock Exchange" which we annotated entirely as an ORG entity, despite models often predicting "New York" as LOC. Therefore, the correct labels are ORG_B for "New" and ORG_I for "York," whereas models might label them as LOC_B and LOC_I, respectively. Commonly used words like "the" which are often tagged as O (Other) may sometimes be present in the name of an organization, such as "The Wall Street Journal". In such cases, we have tagged "The" to be included as part of an ORG entity. In the specific example for "The Wall Street Journal", as shown in Appendix E, we have tagged "The" as ORG_B and the remaining tokens "Wall", "Street", "Journal" as ORG_I tokens. For entities which may represent both an organization and a product, such as "Google", we have only tagged such entities as ORG because our dataset currently does not provide manual annotations for product entities.

F Most Common Entities

Table 4 shows the most common entities within FiNER-ORD. Since the dataset is comprised of financial news articles from July to October 2015, the dataset has a bias towards the news of that time period. An advantage of FiNER-ORD is that it has a unique heterogeneity due to it being composed of English language financial articles published by institutions from around the world unlike the CRA dataset which are from United States Securities and Exchange Commission documents. The most common LOC and ORG entities reflect the global news article source.

Model	learning rate	batch size
BERT-base-cased	1e-5	16
FinBERT-base-cased	1e-5	8
RoBERTa-base	1e-5	8
BERT-large-cased	1e-5	8
RoBERTa-large	1e-5	8

Table 5: Best hyper-parameter configuration for each PLM benchmarked on FiNER-ORD.

G Doccano Annotation Tool

All manual annotation of FiNER-ORD was completed using the open-source Doccano⁷ annotation tool. Figure 2 demonstrates the use of Doccano to manually annotate FiNER-ORD. The output from Doccano contains span-level label information. This information is in the form of a list of lists containing information on the start character, end character, and label of each entity annotated by the manual annotator.

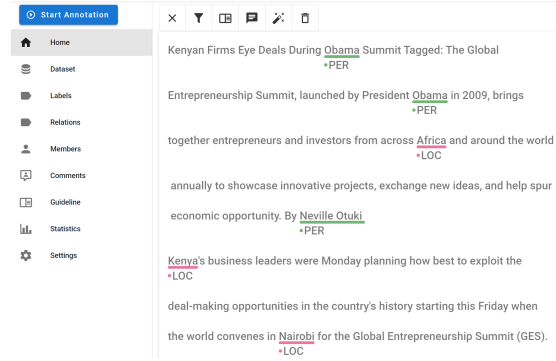


Figure 2: Screenshot of an article in FiNER-ORD manually annotated with the open-source Doccano annotation tool.

H Fine-tuning PLM Details

No pre-training is conducted on the models before proceeding with the fine-tuning process. To determine the most suitable hyper-parameters for each model, we performed a grid search using three different learning rates (1e-4, 1e-5, 1e-6) and three different batch sizes (32, 16, 8). We use a maximum of 100 epochs for training with early stopping criteria. If the validation F1 score doesn't improve by more than or equal to 1e-2 in the next 7 epochs then we use the best model stored earlier as the final 6 fine-tuned model. We report the best hyper-parameter configuration for each PLM in Table 5. All of our experiments are carried out using PyTorch (Paszke et al., 2019) on an NVIDIA RTX

⁷<https://github.com/doccano/doccano>

Train Split	Test Split	PER	LOC	ORG	Weighted Avg.
FiNER	FiNER	0.9263 (0.0025)	0.7717 (0.0152)	0.6769 (0.0130)	0.7648 (0.0057)
CRA	FiNER	0.5918 (0.0868)	0.0145 (0.0020)	0.0411 (0.0174)	0.1730 (0.0262)
CoNLL	FiNER	0.8707 (0.0171)	0.7640 (0.0278)	0.5668 (0.0678)	0.6954 (0.0278)
FiNER+CoNLL	FiNER	0.9107 (0.0185)	0.7992 (0.0071)	0.6441 (0.0183)	0.7522 (0.0141)
FiNER	CoNLL	0.9164 (0.0299)	0.7099 (0.0218)	0.5622 (0.0439)	0.7278 (0.0317)
CRA	CoNLL	0.6498 (0.0562)	0.0492 (0.0173)	0.2078 (0.0685)	0.2988 (0.0316)
CoNLL	CoNLL	0.9553 (0.0080)	0.8974 (0.0214)	0.8038 (0.0180)	0.8849 (0.0146)
FiNER+CoNLL	CoNLL	0.9559 (0.0047)	0.9018 (0.0102)	0.8001 (0.0965)	0.8866 (0.0317)

Table 6: Cross-dataset and combined dataset performance analysis of FiNER, CRA, and CoNLL

Model	Test Split	PER	LOC	ORG	Weighted Avg.
tner/roberta-large-conll2003	FiNER	0.9062	0.8788	0.8004	0.8474
roberta-large-finer	FiNER	0.9384	0.8520	0.8352	0.8637

Table 7: Transfer learning ablation on CoNLL using **tner** framework

A6000 GPU. Each model is initialized with the pre-trained version available in the Hugging Face Transformers library (Wolf et al., 2020). We use three different seeds (5768, 78516, 944601).

I Zero-shot Prompt

"Discard all the previous instructions. Behave like you are an expert named entity identifier. Below a sentence is tokenized and each line contains a word token from the sentence. Identify "Person", "Location", and "Organisation" from them and label them. If the entity is multi token use post-fix _B for the first label and _I for the remaining token labels for that particular entity. The start of the separate entity should always use _B post-fix for the label. If the token doesn't fit in any of those three categories or is not a named entity label it 'Other'. Do not combine words yourself. Use a colon to separate token and label. So the format should be token:label. {sentence}."

J Comparison with CoNLL Dataset

J.1 Combined Dataset Performance

In order to test whether FiNER-ORD and CoNLL can be used together to achieve better NER performance, we combine the training data from both datasets. We test separately on the FiNER-ORD and CoNLL test sets. The Weighted Avg. results shown in Table 6 suggest that when testing on CoNLL, FiNER-ORD can be used to complement CoNLL.

J.2 Training on CoNLL and Testing on FiNER-ORD

Section 3.2 highlighted how financial texts differ from general texts. In particular, financial texts tend

Train Split	Test Split	Percent (%) Overlap
FiNER	FiNER	4.9749
CoNLL	FiNER	1.1462
CoNLL	CoNLL	3.3392
FiNER	CoNLL	0.5927

Table 8: Cross-dataset and intra-dataset analysis of unique 4-gram lexical overlap in FiNER and CoNLL.

to contain a higher ratio of organization tokens and entities compared to person and location tokens and entities. To evaluate how a model trained on the non finance specific CoNLL dataset would do on the FiNER-ORD test split, we evaluate an existing model trained on the CONLL 2003 dataset⁸. The results in Table 7 show that the model only performs better on LOC entities than a similar model trained on the FiNER-ORD train split.

It is important to note that the results presented in Table 7 are not directly comparable to those in Table 3 due to differences in the tokenization methods and input/output formats employed by the models in each table. Specifically, the models in Table 3 produce outputs in the form of lists of tokens and corresponding labels, whereas the models in Table 7 generate outputs consisting of lists of entities along with their respective starting and ending positions.

K Lexical Overlap Analysis

In Table 8, we present a cross-dataset and intra-dataset lexical overlap analysis of unique 4-grams in the train and test splits of FiNER-ORD and CoNLL. We directly follow the percent overlap calculation method presented in Choubey et al. (2023),

⁸<https://huggingface.co/tner/roberta-large-conll2003>

calculating the percentage of a given test split’s unique 4-grams which also exist in a given train split’s set of unique 4-grams. The percent overlap between the train and test splits of FiNER is less than 5%, so lexical similarity is relatively low within the FiNER dataset. The results in Table 8 also indicate that the FiNER test split has more lexical novelty than the CoNLL test split when compared to the CoNLL train split. Furthermore, the FiNER train split and CoNLL test split combination is most lexically different because it has the lowest percentage overlap. Thus, our lexical overlap analysis shows that FiNER is indeed lexically novel when compared to CoNLL, with respect to both the train and test splits of CoNLL, underscoring the value of our contribution with FiNER as a financial domain-specific NER dataset.